Routing, capacity and width in dynamic convolution: VGG-16 (BN)

Replication of the 132-run ResNet-18 program

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Generated from result files by `report/build_model_report.py`

SYNTHETIC TEST DATA. Every number in this document is generated to test the pipeline. None is a result.

This document reports one backbone of the cross-architecture replication. It is generated from the result files: every number, table and figure is computed, and every sentence states what a test returned. Interpretation across backbones is in the cross-model report. Method, statistical protocol and the pre-registered margin are those of the ResNet-18 study and are not repeated here.

1 Setup and coverage

Backbone. VGG-16 (BN), adapted to 32×32 and 64×64 inputs as recorded in `src/models`. Width 64 is the standard network; widths 32 and 16 scale every channel count by $1/2$ and $1/4$. Dynamic layers follow the ResNet-18 rule literally (every $k \times k$ convolution with $k > 1$ inside the blocks; stem, 1×1 convolutions, shortcuts, downsampling layers and head static): here every 3×3 convolution after the first, 12 layers. Initialisation: Kaiming-normal (fan-out) for every static convolution and every bank kernel.

Size. At CIFAR-100 width 64 the static network stores 14.77 M parameters and the $K = 4$ `gap` network 59.29 M, of which 391.8 K are router parameters (Table 14). The bank adds 44.13 M to the static network; the router is 2.7 % of the static network's size. The router is the ResNet-18 design (a two-layer MLP with hidden width $C_{\text{in}}/4$ on the global descriptor), so its size grows with the square of a layer's input channels while a depthwise kernel grows linearly: where dynamic layers are depthwise, the routed arms carry many more parameters than the bank. The `gap−uniform` routing term includes them.

Gate. The first run, static, CIFAR-100, width 64, seed 0, reached 73.01 % against a pre-registered floor of 65.0 %: passed.

Coverage. 130 of 132 runs are present; 2 are missing, of which 1 diverged (non-finite training loss). 0 group(s) ran with gradient accumulation because batch 128 did not fit in memory. Table 1 lists every missing run and every deviation. Paired tests use the seeds both arms have; a verdict computed on fewer than six pairs carries its n.

Table 1: Missing runs, divergences and deviations from the recipe.

dataset	width	K	τ	arm	seed	status
CIFAR-100	32	4	const	`uniform`	2	diverged (non-finite loss)
CIFAR-100	32	2	const	`gap`	1	not run

2 CIFAR-100 at the standard width

Against `static`: `uniform` -0.043 pp (equivalent); `gap` +0.985 pp (higher); `shuffled` -0.045 pp (equivalent); `frozen_random` +0.056 pp (equivalent). 3 of the four arms are equivalent to `static` within 0.5 pp. The largest gap between arm means is 1.030 pp against a median within-arm seed standard deviation of 0.172 pp; one-way ANOVA across the five arms $F = 40.780$, $p = 0.000$. `frozen_random−gap` $= -0.929$ pp ($p = 0.000$, $p_{\text{TOST}} = 0.9969$). Power for `gap−uniform` here: paired sd 0.290 pp, minimum detectable difference 0.305 pp, simulated power 40.1, 91.6 and 100.0 % at true effects of 0.25, 0.5 and 1.0 pp. Table 2 gives the routing entropy by arm.

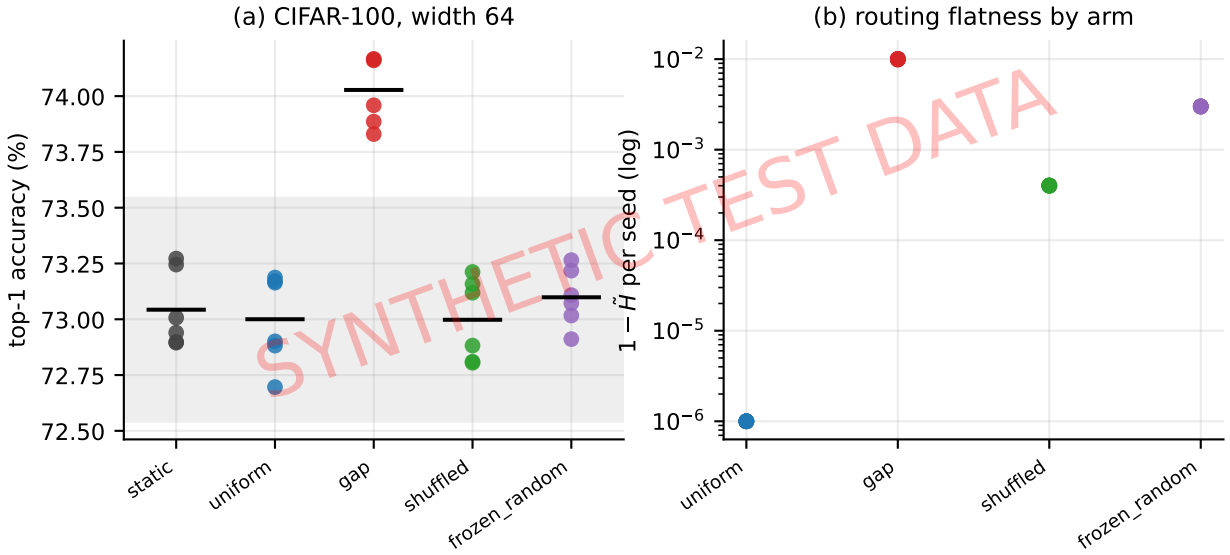


Figure 1: CIFAR-100, width 64. (a) Every run of the five arms; bars are arm means and the shaded band is ±0.5 pp around `static`. (b) Router flatness $1 - \tilde{H}$ per seed, log scale; `uniform` is exactly zero and is drawn at the axis floor.

Table 2: Routing entropy by arm, CIFAR-100 width 64, per-seed means.

arm	mean $\tilde{H}$	min	max	$1 - \tilde{H}$
`uniform`	1.0000	1.0000	1.0000	0.00000
`gap`	0.9900	0.9899	0.9901	0.00999
`shuffled`	0.9996	0.9996	0.9996	0.00040
`frozen_random`	0.9970	0.9970	0.9970	0.00300

3 The capacity sweep on CIFAR-100

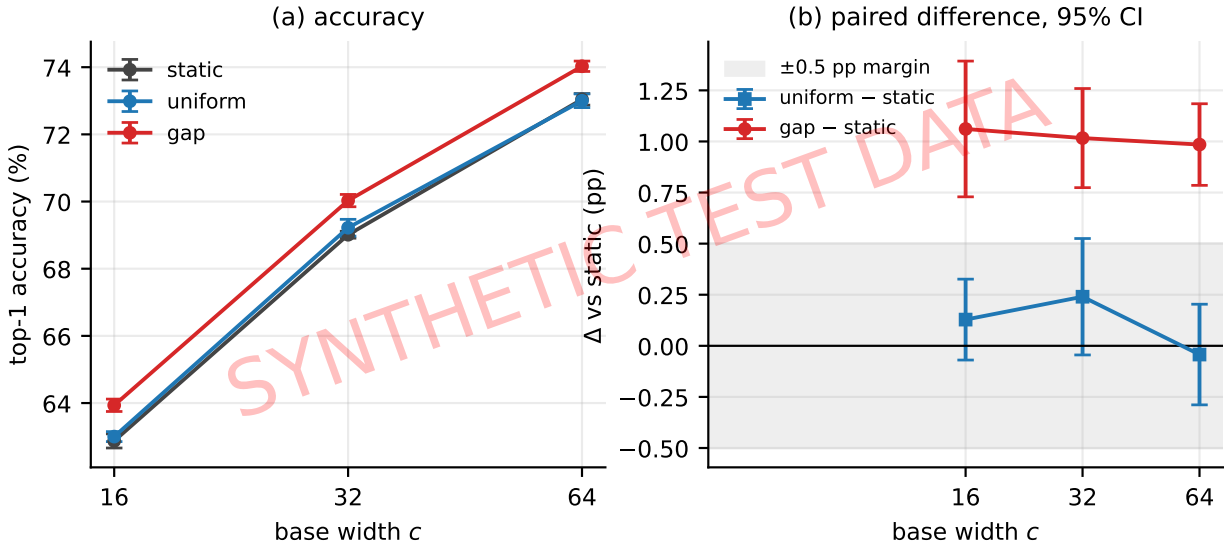


Figure 2: CIFAR-100. (a) Accuracy against base width, ±1 standard deviation over six seeds. (b) Paired difference against `static` with 95 % confidence intervals; the band is the ±0.5 pp margin.

`gap−static`: +1.061 pp at width 16 (higher), +1.017 pp at width 32 (higher), +0.985 pp at width 64 (higher) (Figure 2, Table 13).

4 Bank against routing

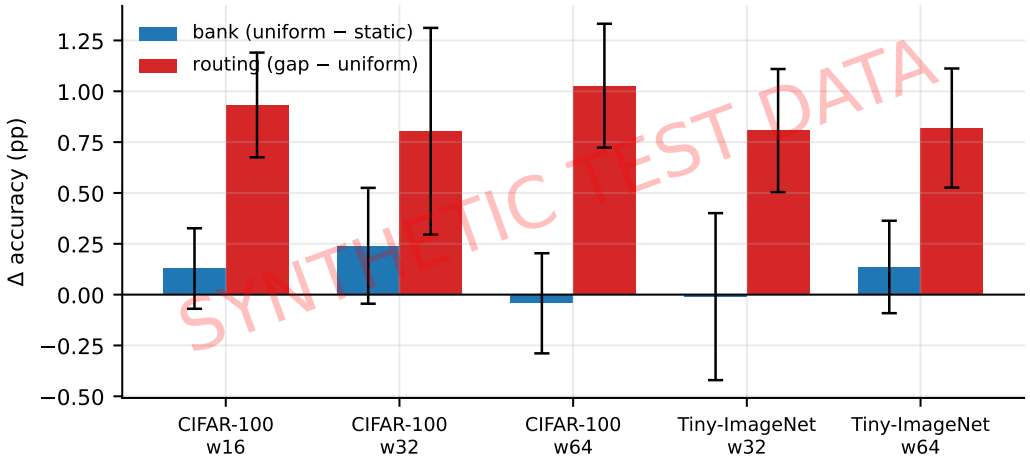


Figure 3: The total effect split into a bank term (`uniform−static`) and a routing term (`gap−uniform`), five conditions, 95 % confidence intervals.

The bank term is equivalent to zero in 5 of 5 conditions; the others: none. The routing term is positive with $p < 0.05$ in 5 of 5 conditions: CIFAR-100 width 16 +0.933 pp ($p = 0.0002$); CIFAR-100 width 32 +0.803 pp ($p = 0.0118$); CIFAR-100 width 64 +1.027 pp ($p = 0.0003$); Tiny-ImageNet width 32 +0.807 pp ($p = 0.0010$); Tiny-ImageNet width 64 +0.819 pp ($p = 0.0008$) (Table 3, Figure 3).

Table 3: Decomposition. Bank verdicts use TOST at ±0.5 pp.

		bank (`uniform−static`)			routing (`gap−uniform`)		total	
dataset	width	Δ (pp)	p_{TOST}	verdict	Δ (pp)	p	Δ (pp)	p
CIFAR-100	16	+0.128	0.0024	equivalent	+0.933	0.0002	+1.061	0.0004
CIFAR-100	32	+0.240	0.0323	equivalent (n=5)	+0.803	0.0118	+1.017	0.0001
CIFAR-100	64	-0.043	0.0025	equivalent	+1.027	0.0003	+0.985	$< 10^{-4}$
Tiny-ImageNet	32	-0.010	0.0139	equivalent	+0.807	0.0010	+0.797	0.0054
Tiny-ImageNet	64	+0.136	0.0046	equivalent	+0.819	0.0008	+0.955	$< 10^{-4}$

5 Bank size

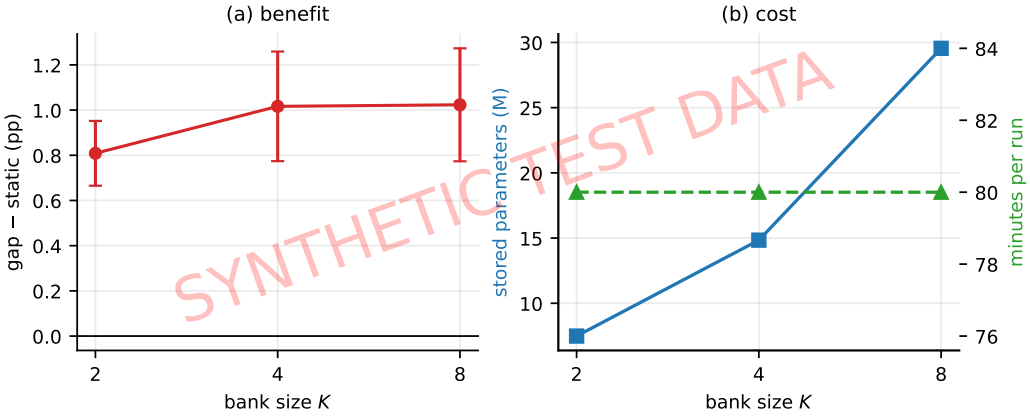


Figure 4: CIFAR-100 width 32, `gap` with $K = 2, 4, 8$. (a) Difference against `static`, 95 % CI. (b) Stored parameters and minutes per run.

$K = 8$ against $K = 2$: +0.144 pp (higher (n=5)) for 3.95× the parameters (Tables 4, 5).

Table 4: Bank size at CIFAR-100 width 32. Differences are against `static`.

K	accuracy	sd	Δ (pp)	95 % CI	p	params (M)	min/run
2	69.824	0.109	+0.809	[+0.665, +0.952]	$< 10^{-4}$	7.48	80.0
4	70.028	0.183	+1.017	[+0.774, +1.259]	0.0001	14.84	80.0
8	70.035	0.208	+1.023	[+0.773, +1.273]	0.0001	29.55	80.0
`static`	69.012	0.103	–	–	–	3.71	20.0

Table 5: Pairwise comparisons between bank sizes.

comparison	Δ (pp)	95 % CI	p	p_{TOST}	verdict
$K = 4$ vs $K = 2$	+0.246	[-0.019, +0.512]	0.0617	0.0284	equivalent (n=5)
$K = 8$ vs $K = 4$	+0.007	[-0.324, +0.338]	0.9597	0.0061	equivalent
$K = 8$ vs $K = 2$	+0.144	[+0.034, +0.255]	0.0221	0.0004	higher (n=5)

6 Tiny-ImageNet

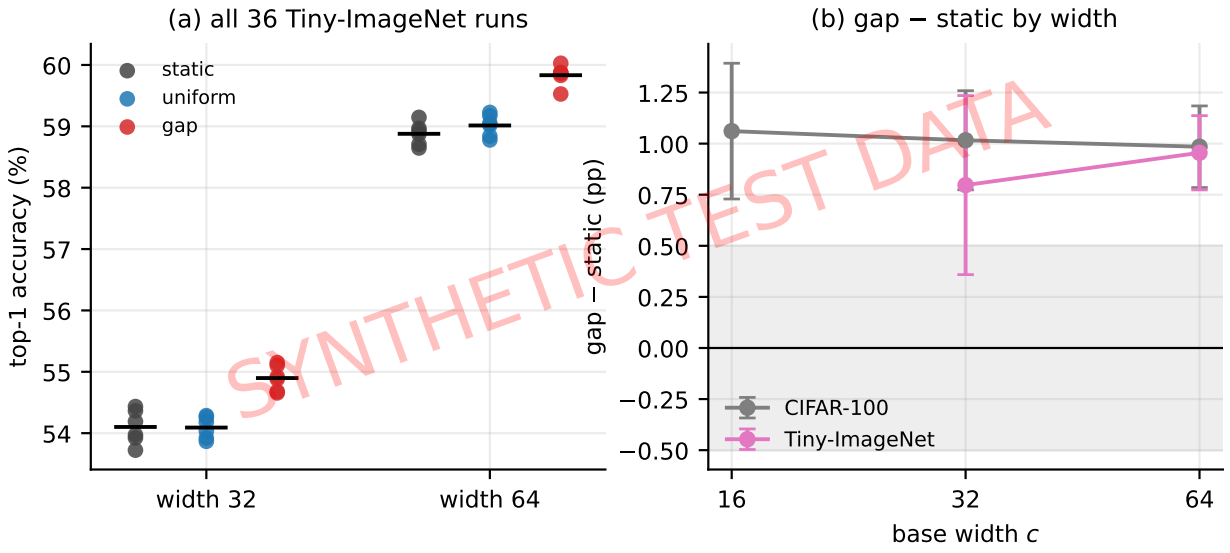


Figure 5: Tiny-ImageNet. (a) Every run at widths 32 and 64. (b) `gap−static` against width on both datasets, 95 % CI; the band is the margin.

`gap−static` is +0.797 pp at width 32 (higher) and +0.955 pp at width 64 (higher).

Across the five conditions the Spearman correlation between the static arm's final training loss and `gap−static` is $\rho = +0.30$ (Table 6, Figure 6).

Table 6: Five conditions ordered by the final training loss of the static arm.

dataset	width	static train loss	`gap − static` (pp)	95 % CI	verdict
CIFAR-100	64	0.0244	+0.985	[+0.785, +1.184]	higher
Tiny-ImageNet	64	0.0306	+0.955	[+0.773, +1.137]	higher
CIFAR-100	32	0.0619	+1.017	[+0.774, +1.259]	higher
Tiny-ImageNet	32	0.0869	+0.797	[+0.359, +1.235]	higher
CIFAR-100	16	0.1119	+1.061	[+0.729, +1.393]	higher

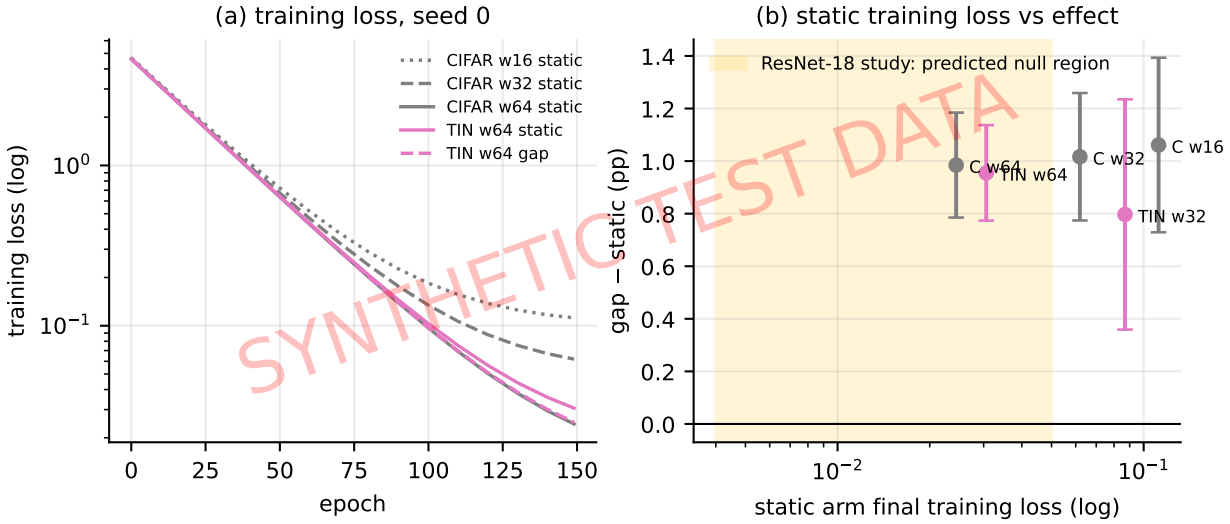


Figure 6: (a) Training loss, seed 0, sampled every ten epochs. (b) `gap−static` against the static arm's final training loss; the shaded region is where the ResNet-18 study predicted a null.

7 Equal parameter budgets

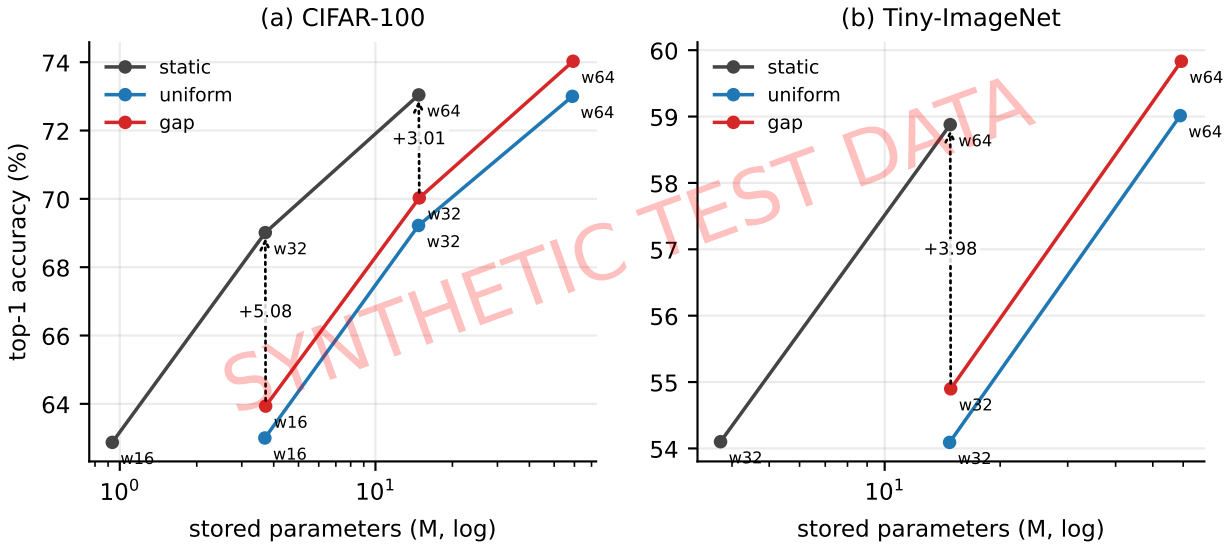


Figure 7: Accuracy against stored parameters. Arrows join `gap` at width c to `static` at width $2c$; the label is the accuracy difference, and the parameter ratio where it is outside $\pm 10\,\%$.

The ResNet-18 comparison of `gap` at width c with `static` at width $2c$ was a matched-budget comparison because a $K = 4$ bank and a doubling of width both multiply the parameters of the dynamic layers by four. That holds only where almost every parameter is in a dynamic layer. Here 3 of 3 pairs are within 10 % and are called matched: CIFAR-100 `gap` w16 vs `static` w32: +5.078 pp, parameter ratio 1.00 (matched); CIFAR-100 `gap` w32 vs `static` w64: +3.015 pp, parameter ratio 1.00 (matched); Tiny-ImageNet `gap` w32 vs `static` w64: +3.980 pp, parameter ratio 1.00 (matched) (Table 7). Table 8 adds the comparison at equal parameters: the static accuracy curve interpolated linearly in log-parameters between the static widths that were run, evaluated at the `gap` network's count. It is an interpolation, not a run, and has no confidence interval.

Table 7: `gap` at width c against `static` at width $2c$. Δ is plain minus routed. The last column is the routed network's parameters divided by the plain one's; a pair is called matched only within 10 %.

dataset	routed	plain	routed	plain	Δ (pp)	95 % CI	p	min/run	param ratio
CIFAR-100	`gap` w16 (3.72M)	`static` w32 (3.71M)	63.934	69.012	+5.078	[+4.858, +5.298]	$< 10^{-4}$	40.0 / 20.0	1.00 (matched)
CIFAR-100	`gap` w32 (14.84M)	`static` w64 (14.77M)	70.028	73.043	+3.015	[+2.701, +3.329]	$< 10^{-4}$	80.0 / 40.0	1.00 (matched)
Tiny-ImageNet	`gap` w32 (14.86M)	`static` w64 (14.82M)	54.898	58.878	+3.980	[+3.802, +4.158]	$< 10^{-4}$	80.0 / 40.0	1.00 (matched)

Table 8: `gap` against the static accuracy curve at equal parameters (interpolated, no CI).

dataset	routed	params (M)	routed acc.	static at equal params	Δ (pp)
CIFAR-100	`gap` w16	3.72	63.934	69.020	+5.087
CIFAR-100	`gap` w32	14.84	70.028	outside the static range	–
Tiny-ImageNet	`gap` w32	14.86	54.898	outside the static range	–

8 Temperature schedule

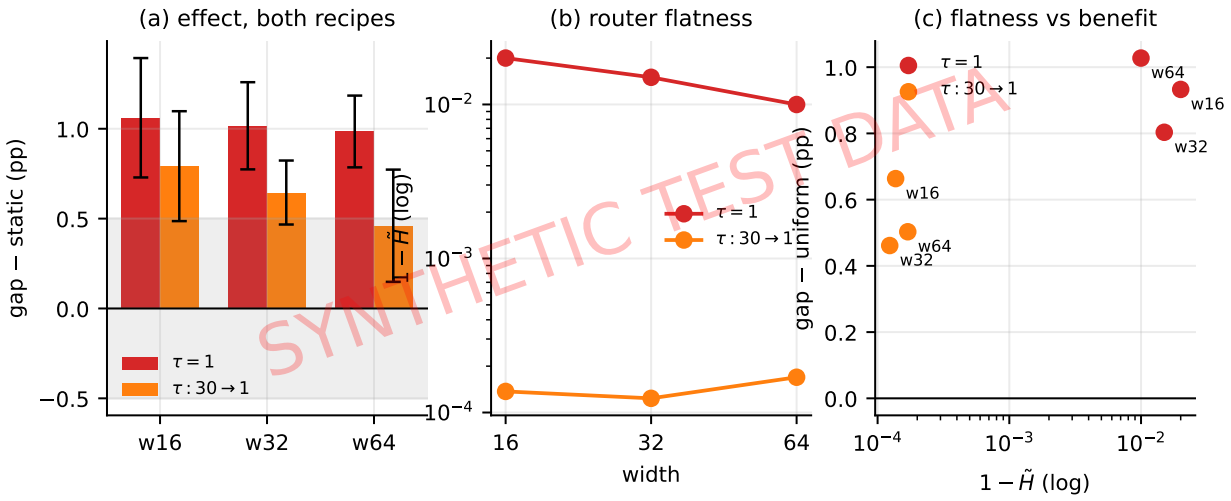


Figure 8: Constant $\tau = 1$ against τ annealed from 30 to 1 over ten epochs, CIFAR-100. (a) `gap−static`. (b) Router flatness. (c) Flatness against benefit over `uniform`.

Annealed minus constant, `gap`: width 16 -0.269 pp (lower); width 32 -0.371 pp (lower); width 64 -0.525 pp (lower) (Table 9).

Table 9: Temperature schedule, CIFAR-100.

	accuracy		annealed − constant		vs `static`			annealed vs `uniform`	
width	$\tau = 1$	annealed	Δ	p	$\tau = 1$	annealed	p_{TOST}	Δ	verdict
16	63.934	63.664	-0.269	0.0246	+1.061	+0.792	0.9712	+0.664	higher
32	70.028	69.657	-0.371	0.0196	+1.017	+0.645	0.9549	+0.461	higher (n=5)
64	74.028	73.503	-0.525	0.0061	+0.985	+0.460	0.3786	+0.503	higher

9 Routing entropy and routing benefit

At width 16 the constant-temperature router's $1 - \tilde{H}$ is 145.9 times the annealed router's; the benefit over `uniform` is +0.933 pp constant and +0.664 pp annealed. Across the 6 width–recipe points the Spearman correlation between $1 - \tilde{H}$ and `gap−uniform` is $\rho = +0.77$ (Table 10).

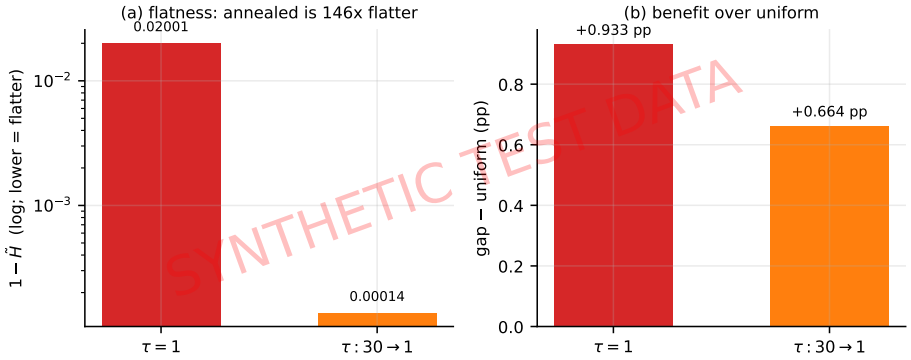


Figure 9: CIFAR-100 width 16, both temperature recipes. (a) Router flatness. (b) Benefit of `gap` over `uniform`.

Table 10: Routing entropy against the benefit over `uniform`.

width	τ	$\tilde{H}$	$1-\tilde{H}$	RMS dev	`gap`−`uniform` (pp)	p
16	1	0.97999	0.02001	0.0589	+0.933	0.0002
16	$30 \to 1$	0.99986	0.00014	0.0049	+0.664	0.0004
32	1	0.98499	0.01501	0.0510	+0.803	0.0118
32	$30 \to 1$	0.99988	0.00012	0.0046	+0.461	0.0128
64	1	0.99001	0.00999	0.0416	+1.027	0.0003
64	$30 \to 1$	0.99983	0.00017	0.0054	+0.503	0.0035

10 Pre-registered predictions

Recorded in `PREREGISTRATION_MODELS.md` on 30 September 2026, before any run of this backbone.

Table 11: Pre-registered predictions and their outcomes.

	prediction	measured	outcome
P1	bank term equivalent to zero in each of 5 conditions	5 of 5	held
P2	static at width $2c$ beats `gap` at width c in each of 3 pairs	3 of 3	held
P3	routing term positive, $p < 0.05$, in at least 3 of 5 conditions	5 of 5	held

11 Complete results

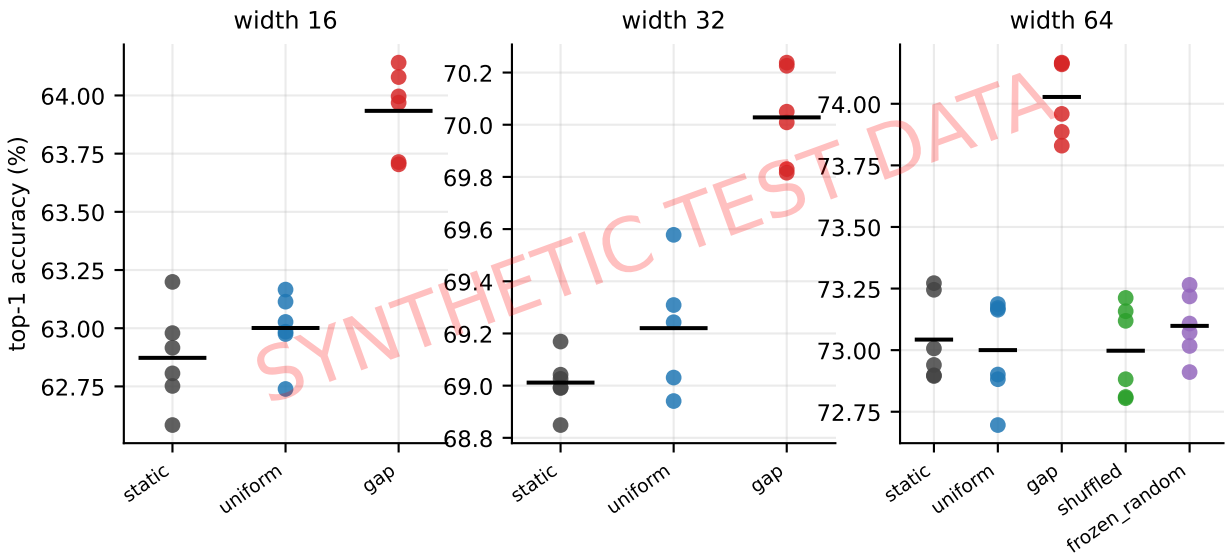


Figure 10: Every CIFAR-100 run at $K = 4$, $\tau = 1$. Bars are arm means.

Table 12: Every run at $K = 4$ and $\tau = 1$: final-epoch top-1 accuracy (%). – marks a missing or diverged run (Table 1).

dataset	width	arm	s0	s1	s2	s3	s4	s5	mean	sd
CIFAR-100	16	`static`	62.75	63.20	62.92	62.58	62.98	62.81	62.873	0.211
CIFAR-100	16	`uniform`	63.11	62.98	63.03	62.74	63.17	62.98	63.001	0.148
CIFAR-100	16	`gap`	63.71	63.70	64.00	63.97	64.08	64.14	63.934	0.185
CIFAR-100	32	`static`	68.99	68.99	69.17	69.03	68.85	69.04	69.012	0.103
CIFAR-100	32	`uniform`	69.58	69.31	–	69.24	68.94	69.03	69.220	0.250
CIFAR-100	32	`gap`	70.01	69.82	70.05	69.83	70.24	70.23	70.028	0.183
CIFAR-100	64	`static`	73.01	73.27	73.25	72.90	72.94	72.89	73.043	0.172
CIFAR-100	64	`uniform`	73.16	73.19	72.90	72.88	72.70	73.17	73.000	0.204
CIFAR-100	64	`gap`	74.16	73.96	74.17	73.83	74.16	73.89	74.028	0.155
CIFAR-100	64	`shuffled`	73.21	73.12	73.16	72.81	72.88	72.81	72.998	0.186
CIFAR-100	64	`frozen_random`	73.27	73.07	72.91	73.11	73.02	73.22	73.099	0.130
Tiny-ImageNet	32	`static`	53.92	53.72	54.43	54.36	54.19	53.97	54.101	0.275
Tiny-ImageNet	32	`uniform`	54.18	54.29	53.92	54.26	53.87	54.03	54.091	0.177
Tiny-ImageNet	32	`gap`	54.66	55.15	54.88	54.68	54.92	55.10	54.898	0.206
Tiny-ImageNet	64	`static`	58.65	59.15	58.71	58.97	58.87	58.93	58.878	0.181
Tiny-ImageNet	64	`uniform`	58.84	59.05	59.17	59.23	58.78	59.02	59.014	0.177
Tiny-ImageNet	64	`gap`	59.83	59.87	59.53	59.87	59.88	60.03	59.833	0.165

Table 13: Paired comparisons against `static`. p_W is the Wilcoxon signed-rank p (floor 0.031 at $n = 6$).

dataset	width	arm	Δ (pp)	95 % CI	p_t	p_W	d_z	p_{TOST}	verdict
CIFAR-100	16	`uniform`	+0.128	[-0.070, +0.326]	0.1565	0.3125	+0.68	0.0024	equivalent
CIFAR-100	16	`gap`	+1.061	[+0.729, +1.393]	0.0004	0.0312	+3.35	0.9963	higher
CIFAR-100	32	`uniform`	+0.240	[-0.045, +0.525]	0.0793	0.1250	+1.05	0.0323	equivalent (n=5)
CIFAR-100	32	`gap`	+1.017	[+0.774, +1.259]	0.0001	0.0312	+4.40	0.9986	higher
CIFAR-100	64	`uniform`	-0.043	[-0.289, +0.204]	0.6746	0.6875	-0.18	0.0025	equivalent
CIFAR-100	64	`gap`	+0.985	[+0.785, +1.184]	$< 10^{-4}$	0.0312	+5.18	0.9992	higher
CIFAR-100	64	`shuffled`	-0.045	[-0.178, +0.088]	0.4217	0.4375	-0.36	0.0002	equivalent
CIFAR-100	64	`frozen_random`	+0.056	[-0.224, +0.335]	0.6301	0.6875	+0.21	0.0047	equivalent
Tiny-ImageNet	32	`uniform`	-0.010	[-0.420, +0.401]	0.9533	1.0000	-0.03	0.0139	equivalent
Tiny-ImageNet	32	`gap`	+0.797	[+0.359, +1.235]	0.0054	0.0312	+1.91	0.9291	higher
Tiny-ImageNet	64	`uniform`	+0.136	[-0.091, +0.363]	0.1845	0.3125	+0.63	0.0046	equivalent
Tiny-ImageNet	64	`gap`	+0.955	[+0.773, +1.137]	$< 10^{-4}$	0.0312	+5.52	0.9993	higher

Table 14: Configuration and cost. Router parameters in thousands; training loss is the final-epoch mean over seeds; minutes are logged wall clock.

dataset	width	arm	params (M)	router (K)	train loss	min/run
CIFAR-100	16	`static`	0.93	0.0	0.1119	10.0
CIFAR-100	16	`uniform`	3.69	0.0	0.1119	40.0
CIFAR-100	16	`gap`	3.72	26.5	0.0819	40.0
CIFAR-100	32	`static`	3.71	0.0	0.0619	20.0
CIFAR-100	32	`uniform`	14.74	0.0	0.0619	80.0
CIFAR-100	32	`gap`	14.84	99.7	0.0469	80.0
CIFAR-100	64	`static`	14.77	0.0	0.0244	40.0
CIFAR-100	64	`uniform`	58.90	0.0	0.0244	160.0
CIFAR-100	64	`gap`	59.29	391.8	0.0206	160.0
CIFAR-100	64	`shuffled`	59.29	391.8	0.0244	160.0
CIFAR-100	64	`frozen_random`	58.90	0.0	0.0244	160.0
Tiny-ImageNet	32	`static`	3.73	0.0	0.0869	20.0
Tiny-ImageNet	32	`uniform`	14.77	0.0	0.0869	80.0
Tiny-ImageNet	32	`gap`	14.86	99.7	0.0644	80.0
Tiny-ImageNet	64	`static`	14.82	0.0	0.0306	40.0
Tiny-ImageNet	64	`uniform`	58.95	0.0	0.0306	160.0
Tiny-ImageNet	64	`gap`	59.34	391.8	0.0250	160.0

Table 15: Every run of this backbone. GPU-hours from logged wall clock.

dataset	width	K	τ	arms	runs	GPU-h
CIFAR-100	16	4	constant	`static`, `uniform`, `gap`	18	9.0
CIFAR-100	32	4	constant	`static`, `uniform`, `gap`	17	16.7
CIFAR-100	64	4	constant	`static`, `uniform`, `gap`, `shuffled`, `frozen_random`	30	68.0
CIFAR-100	32	2	constant	`gap`	5	6.7
CIFAR-100	32	8	constant	`gap`	6	8.0
CIFAR-100	16	4	annealed	`gap`	6	4.0
CIFAR-100	32	4	annealed	`gap`	6	8.0
CIFAR-100	64	4	annealed	`gap`	6	16.0
Tiny-ImageNet	32	4	constant	`static`, `uniform`, `gap`	18	18.0
Tiny-ImageNet	64	4	constant	`static`, `uniform`, `gap`	18	36.0
total					130	190.3